

Tristan Louis

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EDUCATION

Queens College
Bachelor of Science in Computer Science

Graduated: May 2025

EXPERIENCE

Data Science Intern

Dakdan Worldwide- Remote

August 2025 - Present

- Built automated ETL pipelines processing sports-business datasets from **Google Trends**, **Twitter**, and **Twitch**.
- Created **dashboards** and **statistical reports** supporting revenue decisions for marketing and partnerships.
- Automated **data workflows**, reducing manual processing time by **30%**

Software Development Intern

OpenQQuantify - Remote

June 2025 - Present

- Built **backend services** using **Flask**, **FastAPI**, and **SQLAlchemy**, including **REST APIs** and **LLM-integrated endpoints**.
- Designed **modular backend architectures** separating **business logic**, **data access**, and services for scalability.
- Prototyped AI-driven 3D simulation** features with **CesiumJS** and contributed to early **digital twin workflows**.

Data Engineer

Open Avenues Foundation: The Build Fellowship Highlight Reel: Automated Sports Content Generation

September 2024 - December 2024

- Spearheaded the development of an AI-driven sports highlight generator, **leveraging machine learning & computer vision**.
- Engineered a real-time video processing pipeline using **XGBoost** and **OpenCV**, integrating **ball tracking** and **automated transitions**.
- Refined models through iterative testing, **improving highlight accuracy from 67 to 89%** and reducing false positives in play detection.
- Collaborated with engineers and industry mentors, implementing feature engineering techniques to **enhance model performance**.

CLUBS & LEADERSHIP

- Weight Room Club—Active member focused on strength training, fitness, and discipline.
- Queens College Stock Broker Exchange Club—Engaged in financial markets, trading strategies, and investment analysis.
- Queens College CODE FOR ALL—Contributed to coding initiatives promoting accessibility and inclusivity in tech.
- D.I.C.E. (Diversity in Computing and Engineering)—Advocated for diversity and inclusion in STEM fields.

PROJECTS

Predictive Analytics for Diabetes Management

Pandas, scikit-learn, AWS SageMaker, PCA(Principal Component Analysis), StandardScale(scales it to one), mlxtend (Apriori(to learn, association rules), Matplotlib

- Conducted **machine learning-based analysis** of diabetes treatment patterns using **classification models** and **clustering techniques**.
- Applied the **Apriori algorithm** for **association rule mining**, identifying key relationships between **A1C levels**, **medication changes**, and **emergency visits**.
- Developed **classification models** to predict patient outcomes, achieving **85.3% accuracy in medication change predictions** and **90.2% accuracy in emergency visit forecasting**.
- Performed **clustering analysis** to segment patients based on treatment plans, providing insights into personalized diabetes care.

Customer Credit Risk Prediction Using Machine Learning

Python, scikit-learn (Logistic Regression, Decision Tree, Random Forest), SHAP, Streamlit

- Built a **credit risk assessment model** to predict **loan default probabilities** using **XGBoost & Random Forest**.
- Engineered financial features like **debt-to-income ratio**, **credit utilization**, and **payment history trends**, improving accuracy to **88%**.
- Implemented **SHAP (Shapley Additive Explanations)** for explainability, providing financial analysts with insights into **risk factors**.
- Built an **interactive dashboard using Streamlit**, enabling lenders to assess borrower risk in real time and make data-driven loan decisions.

PathMark Interactive BOT

Python, JSON, Flask, Gemini API, SpeechRecognition

- Developed a **voice-activated assistant** that helps users find grocery items by querying a **nested JSON** inventory using natural language.
- Enabled customers to locate items via **keyword or voice input** and receive **audio responses**
- Planning the integration of **NLP** and **voice input** for future versions

TECHNICAL SKILLS

- Programming Languages:** Python, SQL, C++, Java
- Backend:** FastAPI, Flask, REST APIs, SQLAlchemy
- Machine Learning & AI:** XGBoost, Scikit-learn, OpenCV, TensorFlow, Feature Engineering,
- Data Processing & Visualization:** ETL Pipelines, Pandas, NumPy, Streamlit
- Cloud & Deployment:** AWS (S3, Lambda, SageMaker), GCP
- Developer Tools:** Git, Linux, VS Code, PyCharm, IntelliJ

CERTIFICATIONS

- Build Fellowship Completion**—An AI-powered sports highlight generation using ML & computer vision.
- Code Path Intermediate Technical Prep (TIP102)**—Python, data structures, and algorithms
- Udemy Full-Stack Web Development Bootcamp**—Node.js, Express, React, and database management.
- Udemy Python for Data Science & Machine Learning**—Python, data structures, algorithms, and AI applications
- Udemy The complete SQL Bootcamp**—PostgreSQL, window functions, subqueries, and real-world database design